

Sarthak Mohapatra

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SUMMARY

Gen AI Engineer with hands-on experience building production multi-agent systems, LLM-powered pipelines, and scalable data infrastructure. Specialised in LangGraph orchestration, agentic system design, and LLM integration for real-world automation. Currently building an end-to-end agentic data pipeline on Databricks that autonomously generates, validates, and self-heals PySpark workflows.

EXPERIENCE

Software Engineer

July 2025 – Present

Veersa Technologies

New Delhi

- Designed multi-layer (Bronze-Silver-Gold) data pipelines on Databricks for large-scale healthcare datasets, powering downstream ML and analytics workflows.
- Built schema validation and data quality frameworks including null handling, consistency checks, and anomaly detection — directly applicable to LLM data preparation and context retrieval pipelines.
- Engineered batch ETL systems using PySpark with emphasis on scalability, observability, and maintainability in a production healthcare environment.
- Diagnosed and resolved recurring pipeline failures, improving system reliability and reducing client-facing incidents.

Game Data Scientist Intern

Mar 2025 – July 2025

Red Panda Games

- Contributed to a multi-modal AI pipeline integrating LLMs, image generation models, and 3D data workflows — early hands-on experience with cross-modal Gen AI systems.
- Owned end-to-end 3D asset optimization pipeline using headless Blender, enabling automated polygon reduction at scale.
- Built S3-integrated ingestion and processing system with tracking and monitoring for scalable asset workflows.

PROJECTS

End-to-End Agentic Data Pipeline — Databricks

2026 – Present

LangGraph, PySpark, FastAPI, Databricks, Delta Lake, Claude/GPT-4o, NetworkX, PostgreSQL

- Built the planning, orchestration, validation, and deployment framework for an internal agentic data-pipeline platform from a single json.
- Built the **Orchestrator** — Built the orchestration engine coordinating planning, code generation, validation, deployment, and debugging workflows across pipeline stages. (plan → generate → validate → deploy → debug → rerun), with retry logic, partial failure handling, and inter-agent state propagation.
- Designed the **Planner Agent** using Pydantic classes to construct dependency-resolved DAGs from metadata contracts.
- Built the **Spec Service** (FastAPI + Pydantic v2) as the system entry point — validating all incoming pipeline configs before any downstream agent is invoked.
- Engineered **GitHub and Databricks managers** for automated repo syncing and job dispatch, enabling end-to-end CI/CD for generated PySpark scripts across medallion layers.
- Building the **Debugger Agent** — uses LLM to analyse Spark error traces, identify root cause, and generate targeted code patches, minimising MTTR without full pipeline reruns.

Agentic AI Surgical Assistant

Feb 2026

Top 8 — Agentic AI Hackathon

- Designed a multi-agent system with passive, active, and focused operating modes for real-time surgical decision support — handling dynamic context switching under time-critical conditions.
- Built an event-driven pipeline integrating anomaly detection with LLM reasoning, enabling intelligent, context-aware responses to intraoperative events.
- Architected structured inter-agent communication workflows to coordinate autonomous agents without human intervention.

FastAPI ML Inference Gateway

Jan 2026

FastAPI, Scikit-learn, IsolationForest, Pydantic, Joblib

- Built a modular ML inference gateway supporting JSON, CSV, and form-based inputs with dynamic query routing to direct requests to appropriate model endpoints.
- Integrated IsolationForest-based anomaly detection with Pydantic request validation for robust, production-like request handling.
- Designed extensible architecture with serialised model loading via Joblib and multithreaded request handling for scalable inference.

1B Parameter Model Fine-Tuning under 4GB VRAM

Feb 2026

PyTorch, LoRA, HuggingFace Transformers

- Fine-tuned a 1B parameter LLM on consumer-grade hardware using LoRA — achieving a **66% improvement** in embedding quality (15% → 25% cosine similarity) without full model retraining.
- Built a custom evaluation framework to benchmark optimization impact across model checkpoints, enabling data-driven fine-tuning decisions.
- Demonstrated practical understanding of LLM fundamentals: tokenization, embedding spaces, parameter-efficient training, and VRAM constraints.

TECHNICAL SKILLS

- **Gen AI & LLMs:** LangGraph, Multi-Agent Orchestration, Prompt Engineering, Fine-tuning (LoRA), Embeddings, LLM Integration (OLLAMA, Gemini), MCP, Claude code
- **ML & AI:** PyTorch, TensorFlow, HuggingFace Transformers, NLP, Scikit-learn, XGBoost
- **Data Engineering:** PySpark, Databricks, ETL Pipelines, Data Modeling, Apache Spark
- **Backend & Infra:** FastAPI, Flask, Docker, Linux, REST APIs, S3
- **Languages:** Python, SQL

EDUCATION

Maharaja Surajmal Institute of Technology (GGSIPU)

New Delhi

B.Tech in Information Technology

2022 – 2026

CGPA: 8.86/10

ACHIEVEMENTS

- **Top 8** — Agentic AI Hackathon: Built real-time multi-agent decision system under competition constraints
- Competed in **8+ hackathons** focused on AI/ML systems, consistently shipping functional prototypes